SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Malware Analysis Assignment

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	30
Questions	3 (all compulsory)	**Marks per Question**	10
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Faculty	Dr.V.Parthipan	**Slot**	A

Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I ___S.TANISHA(192465008)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student S.TANISHA

Questions :

Exercise 1: Threat Intelligence Platform Investigation

Objective: To investigate suspicious IP addresses, domains, and file hashes using a threat intelligence platform.

Practical Tasks:

- Examine sample IoCs using VirusTotal or another authorized threat intelligence platform.
- Review available reputation scores and threat classifications.
- Record the findings and recommend appropriate security actions.

Expected Output: A threat intelligence investigation report.

Exercise 2: Security Log Collection and Correlation

Objective: To collect and correlate security events from different sources.

Practical Tasks:

- Collect sample authentication, firewall, and web server logs.
- Identify common IP addresses, timestamps, and user activities.
- Correlate events to identify suspicious patterns.

Expected Output: A correlated security event summary.

Exercise 3: Threat Hunting Using Network Logs

Objective: To identify suspicious network activities through proactive threat hunting.

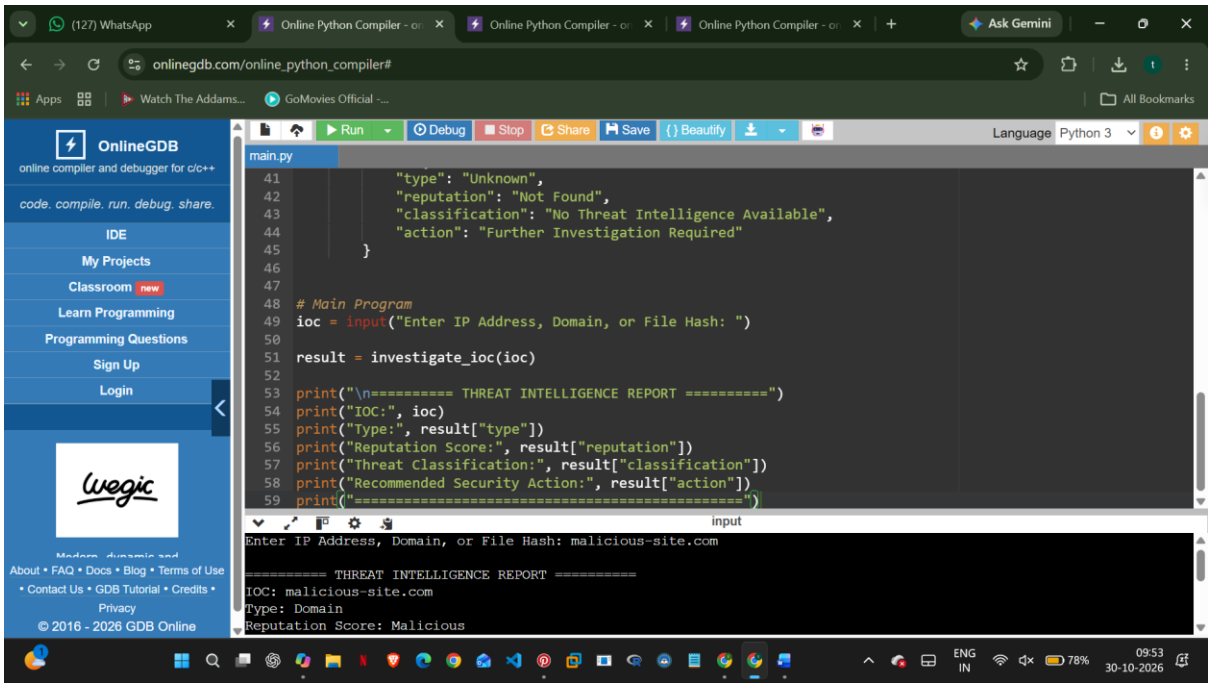
Practical Tasks:

- Examine sample network traffic logs.
- Identify unusual outbound connections and repeated connection attempts.
- Investigate suspicious communication patterns and document relevant IoCs.

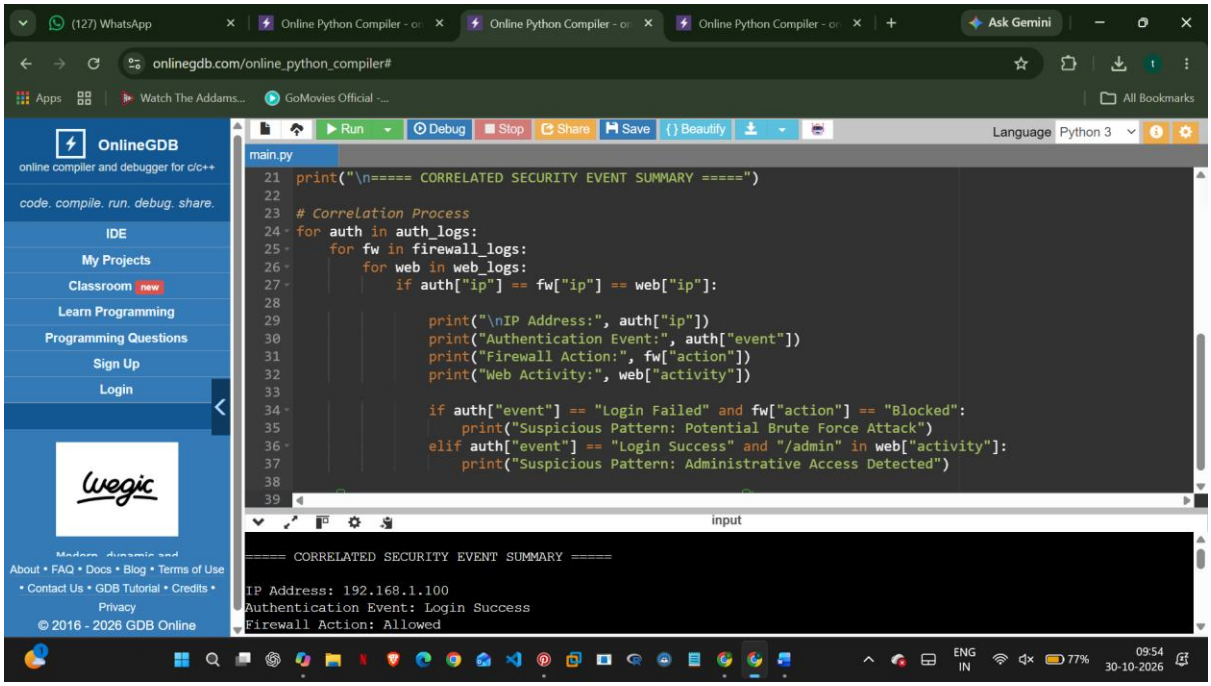
Expected Output: A threat-hunting report with findings and recommendations.

ANSWERS

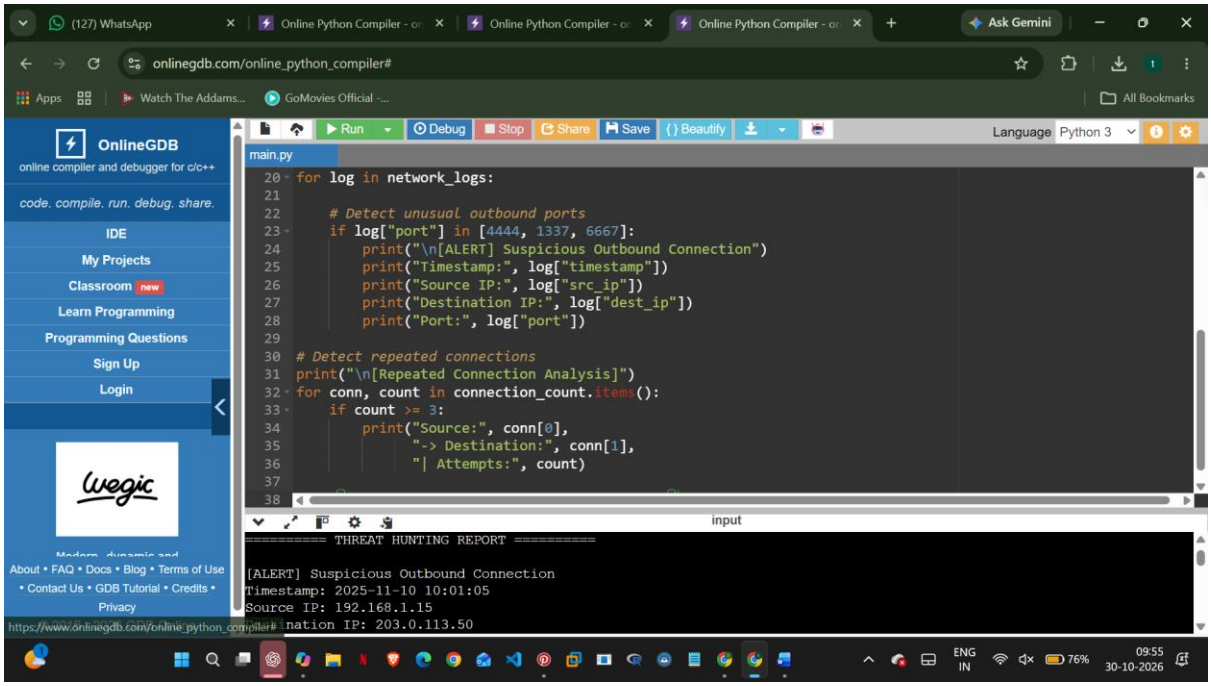
1)


"type": "Unknown",
"reputation": "Not Found",
"classification": "No Threat Intelligence Available",
"action": "Further Investigation Required"
}
Main Program
ioc = input("Enter IP Address, Domain, or File Hash: ")
result = investigate_ioc(ioc)
print("\n========== THREAT INTELLIGENCE REPORT ==========")
print("IOC:", ioc)
print("Type:", result["type"])
print("Reputation Score:", result["reputation"])
print("Threat Classification:", result["classification"])
print("Recommended Security Action:", result["action"])
print("==========")
Enter IP Address, Domain, or File Hash: malicious-site.com
========== THREAT INTELLIGENCE REPORT ==========
IOC: malicious-site.com
Type: Domain
Reputation Score: Malicious

2)


print("\n===== CORRELATED SECURITY EVENT SUMMARY =====")
Correlation Process
for auth in auth_logs:
for fw in firewall_logs:
for web in web_logs:
if auth["ip"] == fw["ip"] == web["ip"]:
print("\nIP Address:", auth["ip"])
print("Authentication Event:", auth["event"])
print("Firewall Action:", fw["action"])
print("Web Activity:", web["activity"])
if auth["event"] == "Login Failed" and fw["action"] == "Blocked":
print("Suspicious Pattern: Potential Brute Force Attack")
elif auth["event"] == "Login Success" and "/admin" in web["activity"]:
print("Suspicious Pattern: Administrative Access Detected")
===== CORRELATED SECURITY EVENT SUMMARY =====
IP Address: 192.168.1.100
Authentication Event: Login Success
Firewall Action: Allowed

3)


onlinegdb.com/online_python_compiler#
OnlineGDB
online compiler and debugger for c/c++
code. compile. run. debug. share.
IDE
My Projects
Classroom new
Learn Programming
Programming Questions
Sign Up
Login
About • FAQ • Docs • Blog • Terms of Use
• Contact Us • GDB Tutorial • Credits •
Privacy
Run
Debug
Stop
Share
Save
{ } Beautify
Language
Python 3
main.py
for log in network_logs:
Detect unusual outbound ports
if log["port"] in [4444, 1337, 6667]:
print("\n[ALERT] Suspicious Outbound Connection")
print("Timestamp:", log["timestamp"])
print("Source IP:", log["src_ip"])
print("Destination IP:", log["dest_ip"])
print("Port:", log["port"])
Detect repeated connections
print("\n[Repeated Connection Analysis]")
for conn, count in connection_count.items():
if count >= 3:
print("Source:", conn[0],
"-> Destination:", conn[1],
"| Attempts:", count)
input
========== THREAT HUNTING REPORT ==========
[ALERT] Suspicious Outbound Connection
Timestamp: 2025-11-10 10:01:05
Source IP: 192.168.1.15
nation IP: 203.0.113.50
https://www.onlinegdb.com/online_python_compiler#
ENG
IN
76%
09:55
30-10-2026